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Student Number

2020 Glenwood High School
YEAR 12-2021
HSC Assessment Task 1

MATHEMATICS EXTENSION 1

**General
Instructions**

Reading time - 5 minutes

Working time - 60 minutes

Write using black pen

Approved calculators may be used.

Show relevant mathematical reasoning and/or calculations

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Total marks: 32

Attempt Questions 1 and 2

Allow about 30 minutes for each question

Start a new writing booklet for each question

Question 1 (18 Marks)

Start a new writing booklet. Your responses should include relevant mathematical reasoning and/or calculations.

- a) Prove by mathematical induction that 3

$$1(1!) + 2(2!) + \dots + n(n!) = (n+1)! - 1$$
for all positive integers
- b) Use the principle of Mathematical induction to prove that $7^n + 2$ is divisible by 3 for all positive integers 3
- c) Solve for x in the interval $0 \leq x \leq 2\pi$, the equation $\sin 2x = \tan x$ 3
- d) $\cos \theta - \sqrt{3} \sin \theta = A \cos(\theta + \alpha)$, By expressing $\cos \theta - \sqrt{3} \sin \theta$ in the form $A \cos(\theta + \alpha)$, solve $\cos \theta - \sqrt{3} \sin \theta = 1$ for $0 \leq \theta \leq 2\pi$. 4
- e) i) Prove the trigonometric identity $\sin 3A = 3 \sin A - 4 \sin^3 A$ 2
ii) Hence show that the equation $6x - 8x^3 = 1$ has the roots $\sin \frac{\pi}{18}$, $\sin \frac{5\pi}{18}$, $\sin \frac{25\pi}{18}$ 2
iii) Hence find the exact value of $\sin \frac{\pi}{18} \times \sin \frac{5\pi}{18} \times \sin \frac{25\pi}{18}$ 1

End of Question 1

Question 2 (14 Marks)

Start a new writing booklet. Your responses should include relevant mathematical reasoning and/or calculations.

- a) Use the principle of Mathematical induction to prove that $3^{3n} + 2^{n+2}$ is divisible by 5 for all positive integers 3
- b) Consider the statement $S(n)$:

$$1 + 5 + 9 + \dots + 4n - 3 = 2n^2 - n$$

and the following attempted proof of this statement by induction.

Proof:

Assume the statement is true for $n = k$. That is,

$$1 + 5 + 9 + \dots + 4k - 3 = 2k^2 - k$$

Next, we shall show it is true for $n = k + 1$ by noting that if

$$1 + 5 + 9 + \dots + 4k - 3 = 2k^2 - k$$

is true, then

$$1 + 5 + 9 + \dots + 4k - 3 + 4(k+1) - 3 = 2(k+1)^2 - (k+1)$$

$$1 + 5 + 9 + \dots + 4k - 3 + 4k + 1 = 2(k+1)^2 - (k+1)$$

Substituting $S(k)$

$$2k^2 - k + 4k + 1 = 2(k+1)^2 - (k+1)$$

$$2k^2 + 3k + 1 = 2(k^2 + 2k + 1) - (k+1)$$

$$2k^2 + 3k + 1 = 2k^2 + 4k + 2 - k - 1$$

Now subtracting $(3k + 1)$ from both sides of this equation

LHS = RHS

The result is true for $n = (k + 1)$ if true for $n = k$

Therefore, by the principle of Mathematical induction, the statement

$P(n)$ is true.

Question 2 continues on next page

- i) Give two reasons why the given proof is incorrect and does not prove $P(n)$. 1
- ii) Provide a correct complete proof by induction of the statement $P(n)$. 2
- c) Use the substitution $t = \tan \frac{\theta}{2}$, to solve the equation 3
- $2 \sin x + \cos x = -1$ for $0 \leq x \leq 2\pi$. Give your answers correct to two decimal places where necessary.
- d) Find the amplitude of the graph $y = 2\sqrt{3} \cos \theta - 2 \sin \theta$ 2
- e) i) Show that $\sin 2x + \sin 4x = \sin 6x$ can be expressed as $\sin 3x \sin 2x \sin x = 0$ 2
- ii) Hence, solve $\sin 2x + \sin 4x = \sin 6x$ for x in the domain $0 \leq \theta \leq \pi$. 1

END OF EXAMINATION

Q1 a)

$$1(1!) + 2(2!) + \dots + n(n!) = (n+1)! - 1$$

Prove the result is true for $n=1$

<u>LHS</u>	<u>RHS</u>
$1(1!) = 1$	$(1+1)! - 1 = 1$

LHS = RHS $\therefore$ true for $n=1$

Assume it's true for $n=k$

$$1(1!) + 2(2!) + \dots + k(k!) = (k+1)! - 1$$

Prove it's true for $n=k+1$

Prove $1(1!) + 2(2!) + \dots + k(k!) + (k+1)(k+1)! = (k+2)! - 1$

<u>LHS</u>
$\underbrace{1(1!) + 2(2!) + \dots + k(k!)}_{S(k)} + (k+1)(k+1)!$

$$(k+1)! - 1 + (k+1)(k+1)!$$

$$(k+1)! [1 + (k+1)] - 1$$

$$(k+1)! (k+2) - 1$$

$$(k+2)! - 1 = \text{RHS}$$

$\therefore$ The result is true for $n=k+1$ if it's true for $n=k$

$\therefore$ by the principle of mathematical induction, the result is true for all positive integers.

Q1 b)

Prove it's true for $n=1$

$$7^1 + 2 = 7 + 2 = 9, \text{ which is divisible by } 3$$

Assume it's true for $n=k$

$$7^k + 2 = 3M \text{ where } M \text{ is an integer.}$$

Prove it's true for $n=k+1$

$$\text{Prove } 7^{k+1} + 2 = 3N, \text{ where } N \text{ is an integer.}$$

$$\text{From assumption, } 7^k = 3M - 2$$

$$7^{k+1} + 2$$

$$= 7^k \times 7 + 2$$

$$= (3M - 2)7 + 2$$

$$= 21M - 14 + 2$$

$$= 21M - 12$$

$$= 3(7M - 4) \text{ is divisible by } 3 \text{ since } M \text{ is an integer}$$

1 c)

$$\sin 2x = \tan x$$

$$0 \leq x \leq 2\pi$$

$$2\sin x \cos x = \frac{\sin x}{\cos x}$$

$$2\sin x \cos^2 x - \sin x = 0$$

$$\sin x (2\cos^2 x - 1) = 0$$

$$\sin x = 0$$

$$x = 0, \pi, 2\pi$$

$$2\cos^2 x - 1 = 0$$

$$\cos x = \pm \frac{1}{\sqrt{2}}$$

$$\text{Base angle} = \frac{\pi}{4}$$

$$x = \frac{\pi}{4}, \pi - \frac{\pi}{4}, \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$= \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$\therefore x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi, \frac{5\pi}{4}, \frac{7\pi}{4}, 2\pi$$

1 d)

$$\cos \theta - \sqrt{3} \sin \theta = R \cos (\theta + \alpha)$$

$$\cos \theta - \sqrt{3} \sin \theta = R [\cos \theta \cos \alpha - \sin \theta \sin \alpha]$$

$$\cos \theta - \sqrt{3} \sin \theta = R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$\therefore R \cos \alpha = 1 \quad \text{--- (1)}$$

$$R \sin \alpha = \sqrt{3} \quad \text{--- (2)}$$

$$\textcircled{1}^2 + \textcircled{2}^2$$

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 1 + 3$$

$$R^2 = 4$$

$$R = 2$$

$$\textcircled{1} \div \textcircled{2}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$= \frac{\pi}{6}$$

$$\therefore \cos \theta - \sqrt{3} \sin \theta = 2 \cos \left(\theta + \frac{\pi}{6} \right)$$

$$\cos \theta - \sqrt{3} \sin \theta = 1$$

$$2 \cos \left(\theta + \frac{\pi}{6} \right) = 1 \quad 0 \leq \theta \leq 2\pi$$

$$\cos \left(\theta + \frac{\pi}{6} \right) = \frac{1}{2} \quad \frac{\pi}{6} \leq \theta + \frac{\pi}{6} \leq 2\pi + \frac{\pi}{6}$$

$$\text{Base angle} = \frac{\pi}{3}$$

$$\theta + \pi/6 = \frac{\pi}{3}, 2\pi - \frac{\pi}{3},$$

$$\theta + \pi/6 = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$\theta = \frac{\pi}{3} - \frac{\pi}{6}, \frac{5\pi}{3} - \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}, \frac{3\pi}{2}$$

Q1) e)

$$\sin 3A = \sin (2A + A)$$

$$= \sin 2A \cos A + \cos 2A \sin A$$

$$= 2 \sin A \cos A \cos A + (1 - 2 \sin^2 A) \sin A$$

$$= 2 \sin A \cos^2 A + \sin A - 2 \sin^3 A$$

$$= 2 \sin A (1 - \sin^2 A) + \sin A - 2 \sin^3 A$$

$$= 2 \sin A - 2 \sin^3 A + \sin A - 2 \sin^3 A$$

$$= 3 \sin A - 4 \sin^3 A$$

b)

$$6x - 8x^3 = 1$$

$$2(3x - 4x^3) = 1$$

$$\text{Let } x = \sin x$$

$$2(3 \sin A - 4 \sin^3 A) = 1$$

from ①

$$3 \sin A - 4 \sin^3 A = \sin 3A$$

$$\therefore 2 (\sin 3A) = 1$$

$$\sin 3A = \frac{1}{2}$$

$$\text{Base angle} = \frac{\pi}{6}$$

$$3A = \frac{\pi}{6}, \pi - \frac{\pi}{6}, 2\pi + \frac{\pi}{6}, \dots$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \dots$$

$$A = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \dots$$

The solutions are $\sin \frac{\pi}{18}, \sin \frac{5\pi}{18}, \sin \frac{13\pi}{18}, \sin \frac{17\pi}{18}, \sin \frac{25\pi}{18}$

Distinct roots are $\sin \frac{\pi}{18}, \sin \frac{5\pi}{18} \times \sin \frac{25\pi}{18}$

$$c) 8x^3 - 6x + 1 = 0$$

$$\alpha\beta\gamma = -\frac{d}{a} = -\frac{1}{8}$$

$$\therefore \sin \frac{\pi}{18} \times \sin \frac{5\pi}{18} \times \sin \frac{25\pi}{18} = -\frac{1}{8}$$

Q2 a)

Prove it's true for $n=1$

$$3^{3(1)} + 2^{(1)+2}$$

$$= 3^3 + 2^3$$

$$= 27 + 8$$

$$= 35, \text{ which is divisible by } 5$$

Assume it's true for $n=k$, where k is an integer.

$$3^{3k} + 2^{k+2} = 5M \text{ where } M \text{ is an integer.}$$

Prove it's true for $n=k+1$

$$\therefore \text{Prove } 3^{3(k+1)} + 2^{(k+1)+2} = 5N$$

$$\underline{\text{LHS}} \quad 3^{3k} \times 3^3 + 2^k \times 2^3$$

$$\text{from assumption } 3^{3k} = 5M - 2^{k+2}$$

$$3^3(5M - 2^{k+2}) + 2^k \times 2^3$$

$$3^3 \times 5M - 3^3 \times 2^k \times 2^2 + 2^3 \times 2^k$$

$$3^3 \times 5M - 108 \times 2^k + 8 \times 2^k$$

$$3^3 \times 5M - 100 \times 2^k$$

$$5(27M - 20 \times 2^k) \text{ which is divisible by } 5$$

since $M \times k$ are integers

$\therefore$ The result is true for $n=k+1$ if true for $n=k$
By the principle of Mathematical Induction the result
is true for all positive integers.

Q2 b)

a) $n=1$ is not proven

when proving for $n=k+1$, rather than LHS & RHS solved separately. They are done side by side and $(3k+1)$ is subtracted from both sides.

b). $1 + 5 + 9 + \dots + 4n - 3 = 2n^2 - n$

Prove it's true for $n=1$

$$\text{LHS} = 1$$

$$\text{RHS} = 2(1)^2 - 1 = 2 - 1 = 1$$

$\therefore \text{LHS} = \text{RHS}$ the result is true for $n=1$

Assume the result is true for $n=k$

$$\therefore \text{Assume } 1 + 5 + 9 + \dots + 4k - 3 = 2k^2 - k$$

Prove the result is true for $n=k+1$

$$\text{Prove } 1 + 5 + 9 + \dots + 4k + 3 + 4(k+1) - 3 = 2(k+1)^2 - (k+1)$$

LHS

$$\underbrace{1 + 5 + 9 + \dots + 4k + 3 + 4(k+1) - 3}_{S(k)}$$

$$= 2k^2 - k + 4(k+1) - 3$$

$$= 2k^2 - k + 4k + 4 - 3$$

$$= 2k^2 + 4k + 2 - k + 1$$

$$= 2(k+1)^2 - (k+1)$$

$$= \text{RHS}$$

$\therefore$ Result is true for $n = k+1$ if it's true for $n = k$

$\therefore$ by the principle of mathematical induction it's true for $n \geq 1$

Q2 c)

$$2 \sin x + \cos x = -1$$

$$2 \left(\frac{2t}{1+t^2} \right) + \frac{1-t^2}{1+t^2} = -1$$

$$\frac{4t + 1 - t^2}{1+t^2} = -1$$

$$4t + 1 - t^2 = -1 - t^2$$

$$4t + 1 - \cancel{t^2} + 1 + \cancel{t^2} = 0$$

$$4t + 2 = 0$$

$$t = -\frac{1}{2}$$

$$\tan \frac{x}{2} = -\frac{1}{2}$$

$$0 \leq x \leq 2\pi$$

$$\frac{x}{2} = \pi - 0.4636$$

$$0 \leq x/2 \leq \pi$$

$$\frac{x}{2} = 2.6779$$

$$x = 5.36$$

Test $x = \pi$

$$2 \sin \pi + \cos \pi = 2 \times 0 + (-1) = -1$$

$\therefore x = \pi$ is a solution

$$\therefore x = 5.36, \pi$$

Q2 a)

$$y = 2\sqrt{3} \cos \theta - 2 \sin \theta$$

$$2\sqrt{3} \cos \theta - 2 \sin \theta = R \cos(\alpha + \beta)$$

$$\alpha = \tan^{-1}(b/a) = \tan^{-1}(2/2\sqrt{3})$$
$$= \tan^{-1}(1/\sqrt{3})$$
$$= \pi/6$$

Note:-
could skip this

$$R = \sqrt{(2\sqrt{3})^2 + (-2)^2}$$

$$= \sqrt{12 + 4}$$

$$= \sqrt{16} = 4$$

$$\therefore y = 4 \cos(\alpha + \pi/6)$$

$$\therefore \text{Amplitude} = 4$$

20)

$$\sin 2x + \sin 4x = \sin 6x$$

LHS

$$A+B = 4x$$

$$A-B = 2x$$

$$2A = 6x$$

$$A = 3x$$

$$B = 4x - 3x = x$$

$$\therefore \sin 2x + \sin 4x = 2 \sin 3x \cos x$$

$$\therefore 2 \sin 3x \cos x = \sin 6x$$

$$2 \sin 3x \cos x = 2 \sin 3x \cos 3x$$

$$2 \sin 3x \cos x - 2 \sin 3x \cos 3x = 0$$

$$2 \sin 3x (\cos x - \cos 3x) = 0$$

$$2 \sin 3x \times 2 \sin x \sin 2x = 0$$

$$4 \sin 3x \sin x \sin 2x = 0$$

$$\sin 3x \sin 2x \sin x = 0$$

$$3x = 0, \pi, 2\pi, 3\pi$$

$$2x = 0, \pi, 2\pi$$

$$x = 0, \pi$$

$$x = 0, \pi/3, \pi, \pi/2$$

$$A+B = 3x$$

$$A-B = x$$

$$2A = 4x$$

$$A = 2x$$

$$B = 3x - 2x = x$$

